

PORTOFOLIO

STUDI KASUS

Pemeliharaan Prediktif Berbasis LSTM untuk Pipa Minyak & Gas

1

PEMANGGILAN DATA

```
Mounted at /content/drive
Pipe_Size_mm Thickness_mm Material Grade \
0 800 15.48 Carbon Steel ASTM A333 Grade 6
1 800 22.00 PVC ASTM A106 Grade B
2 400 12.05 Carbon Steel API 5L X52
3 1500 38.72 Carbon Steel API 5L X42
4 1500 24.32 HDPE API 5L X65

Max_Pressure_psi Temperature_C Corrosion_Impact_Percent \
0 300 84.9 16.04
1 150 14.1 7.38
2 2500 0.6 2.12
3 1500 52.7 5.58
4 1500 11.7 12.29

Thickness_Loss_mm Material_Loss_Percent Time_Years Condition
0 4.91 31.72 2 Moderate
1 7.32 33.27 4 Critical
2 6.32 52.45 7 Critical
3 6.20 16.01 19 Critical
4 8.58 35.28 20 Critical
```

2

PREPROCESSING DATA

```
Pipe_Size_mm 0
Thickness_mm 0
Material 0
Thickness_Loss_mm 0
Material_Loss_Percent 0
Time_Years 0
Condition 0
dtype: int64

Pipe_Size_mm Thickness_mm Thickness_Loss_mm Material_Loss_Percent \
0 800 15.48 4.91 31.72
1 800 22.00 7.32 33.27
2 400 12.05 6.32 52.45
3 1500 38.72 6.20 16.01
4 1500 24.32 8.58 35.28

Time_Years Material_Carbon Steel Material_Fiberglass Material_HDPE \
0 2 True False False
1 4 False False False
2 7 True False False
3 19 True False False
4 20 False False True

Material_PVC Material_Stainless Steel Condition_Critical \
0 False False False
1 True False True
2 False False True
3 False False True
4 False False True
```

4

NORMALISASI DATA

DataFrame after Normalization:

	Pipe_Size_mm	Thickness_mm	Thickness_Loss_mm	Material_Loss_Percent	Time_Years
0	0.626858	-0.056296	4.91	-0.322624	-1.536785
1	0.626858	0.562119	7.32	-0.289347	-1.256375
2	-0.275096	-0.381627	6.32	0.122424	-0.835761
3	2.205277	2.147993	6.20	-0.659898	0.846697
4	2.205277	0.782169	8.58	-0.246195	0.986901

5

MEMBANGUN MODEL

```
LinearRegression
LinearRegression()
```

7

PERBANDINGAN MODEL

Linear Regression Model Evaluation:
Mean Absolute Error (MAE): 0.8699272155810707
Mean Squared Error (MSE): 1.1367285117327666
R-squared (R2): 0.870342136090876

LSTM Model Evaluation (from previous step):
Mean Absolute Error (MAE): 0.3057489394495884
Mean Squared Error (MSE): 0.1793241244705918
R-squared (R2): 0.9795458786453869

Model Comparison:
LSTM model shows better R-squared performance compared to Linear Regression.

10/10 — 0s 22ms/step

LSTM Model Evaluation with Normalized Data:
Mean Absolute Error (MAE): 0.3057489394495884
Mean Squared Error (MSE): 0.1793241244705918
R-squared (R2): 0.9795458786453869

6

EVALUASI PERFORMA MODEL

8

VISUALISASI DASHBOARD

